

week 42 MPGs MSA

focus on the writing and also prepare the presentation

1. presentation. not too much tech details, the big picture... summary as a plot next week? outline 4 parts... 1. meta 2. ps 3. tailocin (TFA HTF) tape measure 4. HPA
2. write the introduction...
3. JC paper

Tuesday:

tonight write HPA ATR1... seq hotspot... RPP1 to albugo...

Wednesday:

slides?

Thursday:

Ps, tailocin tail fiber...

Friday:

slides and tape measure...

Weekend:

JC, relax, research question, gap and all finish...

Monday:

JC, slides...

Tuesday:

do presentation, share thesis intro

prepare also the progress update/ upgrade report...

intro 2000 + future...

ppt final04_12_2024_updates_ATR1_tailocin

More characterization of At and microbial populations:

Done:

Ancient DNA authentication

Allelic polymorphism

Tree topology by r2t orthologs and SNPs

PCA

Future:

fineSTRUCTURE, clustering Pop substructure, also c) fineSTRUCTURE coancestry matrix. Chunk count is the number of genomic segments donated by isolates in rows to isolates in columns (in log scale). Darker colours indicate higher relatedness.

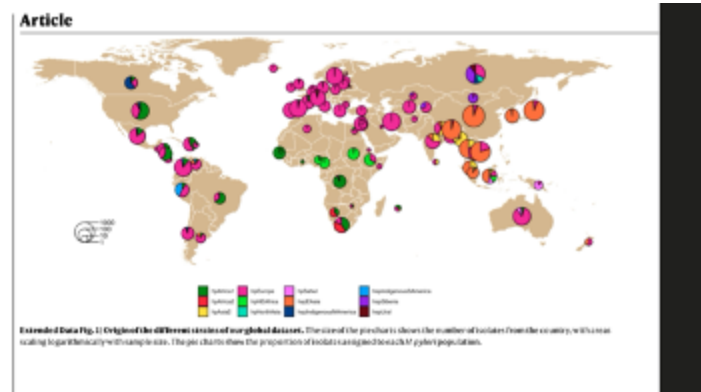
Ref: <https://www.biorxiv.org/content/10.1101/2024.10.24.619980v1.full>

Recophy, pairwise diversity profile... Ps, HPA

Relicts: ancestral alleles, tree...

RPP1, O-antigen

Also the origin map <https://www.nature.com/articles/s41586-024-07991-z>



<https://www.biorxiv.org/content/10.1101/2024.12.03.626388v1>

New paper describing *P. viridiflava* in *Marchantia*, a different lineage of plants
six fasta:

232580E_PG20.fasta 248141E_PG1.fasta 248143E_PG14.fasta

232581E_PG32.fasta 248142E_PG11.fasta 248144E_PG39.fasta

we have all of them short version / two deletions of 123bp except
248141E_PG1.fasta which has no type3 effector in the paper

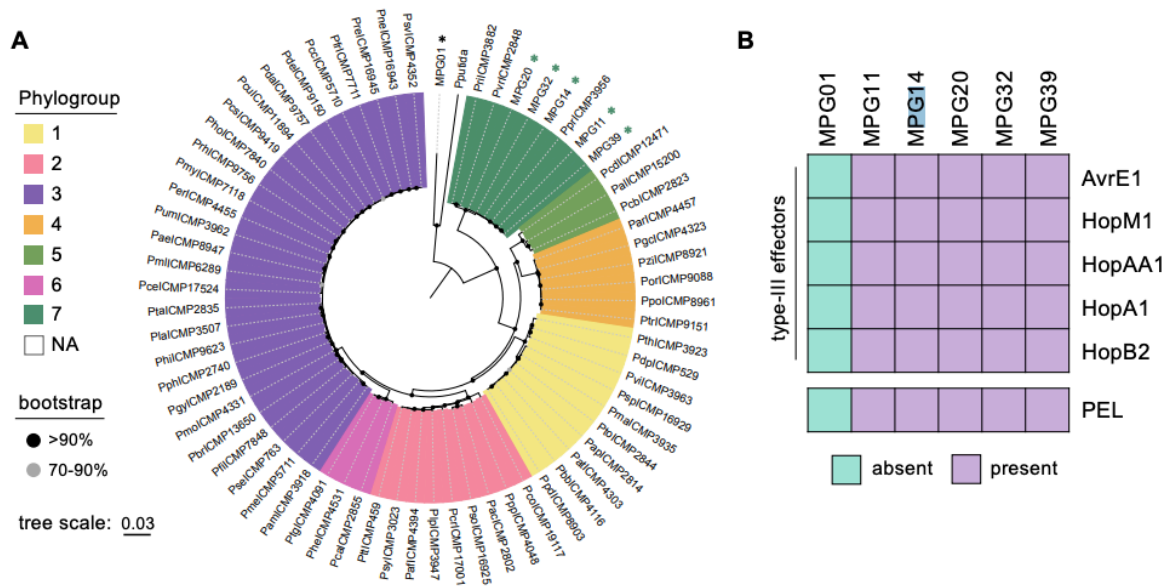


Figure 2. Liverwort-associated *Pseudomonas* belong to phylogroup 7 of the *P. syringae* species complex

(A) Maximum likelihood phylogeny of concatenated single-copy orthologs shared between MPG isolates, diverse *P. syringae* species complex members, and a *P. putida* outgroup (genomes listed in Table S1). Different colours represent different phylogroups of the species complex. Tree scale = substitutions/site.

(B) Presence or absence of type-III effectors or pectate lyase (PEL) virulence factors in liverwort-associated MPG isolate genomes.

A maximum likelihood species phylogeny of 706 concatenated orthologs shared between each MPG isolate and diverse representatives from across the *P. syringae* complex revealed that virulent MPG isolates were situated within phylogroup (PG) 7 alongside *P. viridiflava*, whereas non-infectious MPG01 was excluded from the complex similar to a *P. putida* outgroup (Fig. 2A).

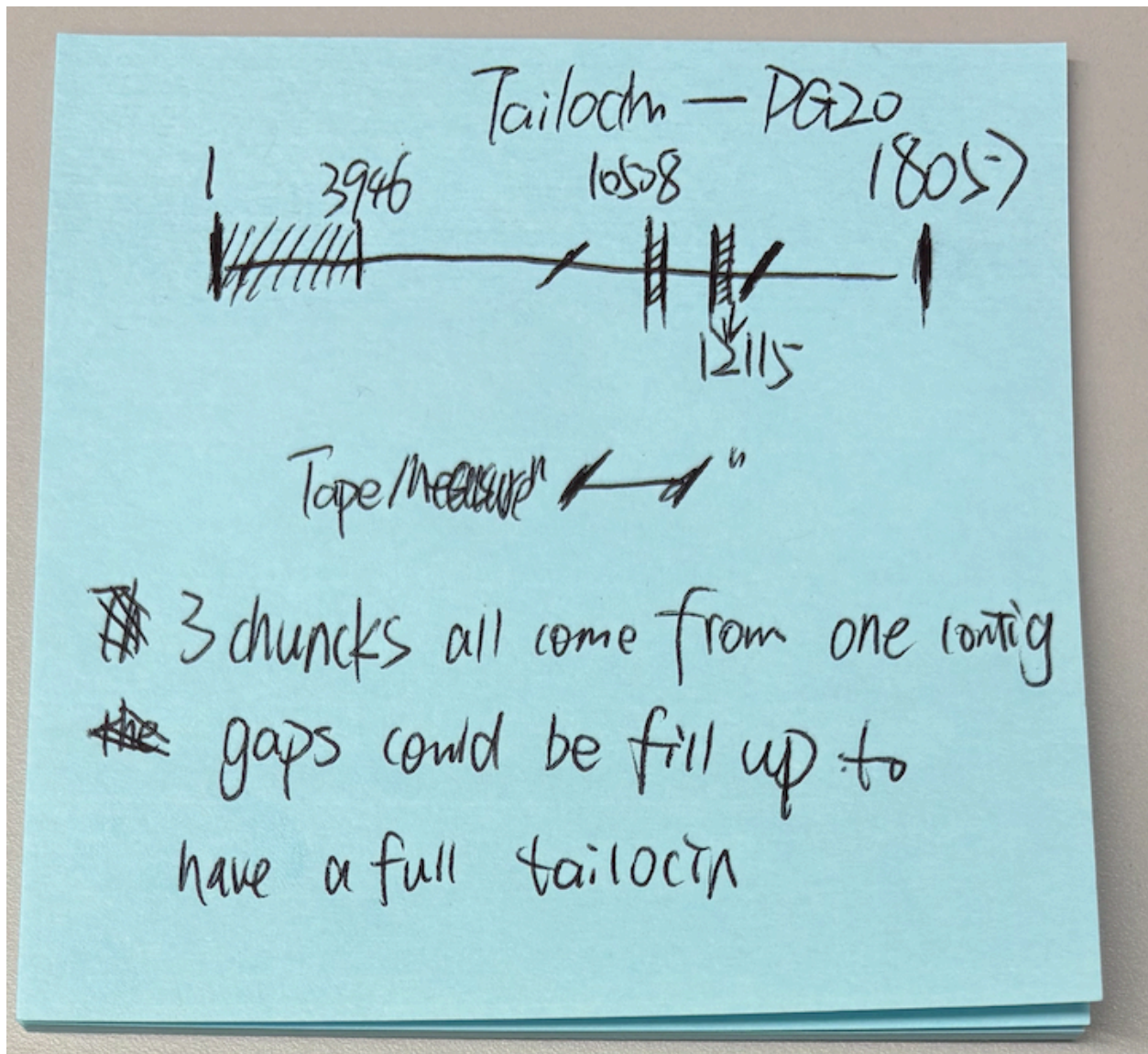
minimap example: PG20

```
tailocin    18057  3946  10508  -   NODE_18_length_85793_cov_33.09174
0   85793  61109  67500  6224  6562  60   NM:i:338    ms:i:3047
AS:i:2718   nn:i:0  tp:A:P  cm:i:441  s1:i:4936  s2:i:0  de:f:0.0264  zd:
```


i:2 rl:i:0 cg:Z:92M123I1369M48I4930M

tailocin 18057 12115 18057 - NODE_18_length_85793_cov_33.09174
0 85793 53683 59625 5773 5942 60 NM:i:169 ms:i:2562
AS:i:2562 nn:i:0 tp:A:P cm:i:365 s1:i:4085 s2:i:0 de:f:0.0284 z
d:i:1 rl:i:0 cg:Z:5942M

tailocin 18057 10819 12060 - NODE_18_length_85793_cov_33.09174
0 85793 59680 60921 1203 1241 60 NM:i:38 ms:i:481 AS:i:48
1 nn:i:0 tp:A:P cm:i:56 s1:i:627 s2:i:0 de:f:0.0306 zd:i:3 rl:i:0 cg:
Z:1241M



PG20:

$$|10508-10819|=311$$

$$|60921-61109|=188$$

$$311-188=123$$

and also cg:Z:92M**123**I1369M48I4930M

so two 123 bps

about tail fiber:

```

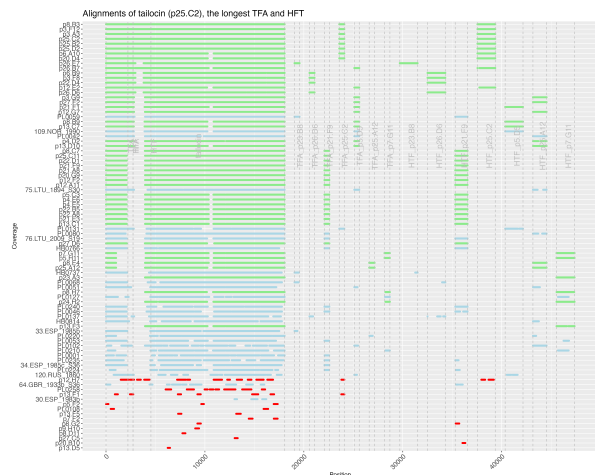
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tailocin 18857 12115 18857 - NODE_18_length_85793_cov_33.891740 85793 53683 59625 5773 5921 60
tailocin 18857 18819 12868 - NODE_18_length_85793_cov_33.891740 85793 59688 68921 1283 1241 60
TFA_p7.G11 544 84 544 - NODE_18_length_85793_cov_33.891740 85793 68751 69233 485 442 60
HTF_p25.F9 1245 639 1245 - NODE_18_length_85793_cov_33.891740 85793 66904 67538 588 686 60
HTF_p25.C2 1883 1287 1883 - NODE_18_length_85793_cov_33.891740 85793 66904 67588 576 596 60
HTF_p5.D5 1883 1193 1883 - NODE_18_length_85793_cov_33.891740 85793 66904 67514 591 618 60
HTF_p25.A12 1383 965 1383 - NODE_18_length_85793_cov_33.891740 85793 66904 67722 798 818 60
HTF_p7.G11 1838 9 1838 - NODE_18_length_85793_cov_33.891740 85793 66904 68725 1788 1823 60

```

```

264      zd:i:2  rl:i:0  cg:Z:92M123I1369M48I4930M
284      zd:i:1  rl:i:0  cg:Z:5942M
rl:i:0    cg:Z:1241M
cg:Z:462M
cg:Z:606M
cg:Z:596M
cg:Z:610M
cg:Z:818M
rl:i:0    cg:Z:1821M

```



HTF and TFA are located around 3000-4000...

in the five new Pv, we dont have these info but some haplotypes of TFA and HTF mapped. For instance, PG20 has a TFA_p7.G11 partially mapped (84/546 unmapped) and a HTF_p7.G11 nearly full mapped (9/1830 unmapped).

PG11:

```

tailocin 18057 3946 10539 - NODE_1_length_588386_cov_32.55114
7 588386 563320 569742 6243 6593 60 NM:i:350 ms:i:2869
AS:i:2509 nn:i:0 tp:A:P cm:i:433 s1:i:4909 s2:i:0 de:f:0.0282 z
d:i:2 rl:i:0 cg:Z:123M123I1369M48I4930M
tailocin 18057 12115 18057 - NODE_1_length_588386_cov_32.55114
7 588386 555925 561867 5771 5942 60 NM:i:171 ms:i:2522
AS:i:2522 nn:i:0 tp:A:P cm:i:365 s1:i:4139 s2:i:0 de:f:0.0288 zd:
i:1 rl:i:0 cg:Z:5942M
tailocin 18057 10819 12060 - NODE_1_length_588386_cov_32.55114
7 588386 561922 563163 1203 1241 60 NM:i:38 ms:i:481 AS:i:
481 nn:i:0 tp:A:P cm:i:58 s1:i:658 s2:i:0 de:f:0.0306 zd:i:3 rl:i:0 c
g:Z:1241M

```

|10539-10819|=280

|563320-563163|=157

280-157=123

cg:Z:123M123I1369M48I4930M

#we have a 123 insertion (deletion since we treat taillocin p25.C2 as the query and each isoalte as the ref when using minimap) in every strand '-' paf except >p12.G7_tape so please add the 2526 full p12.G7 back manually

also two 123 bp deletions

tail fiber:

```
taillocin 18057 3946 18059 - NODE_1_length_588386_cov_32.551147 588386 563328 569742 6243 6593 68
taillocin 18057 12115 18057 - NODE_1_length_588386_cov_32.551147 588386 565925 561867 5771 6942 68
taillocin 18057 18819 12868 - NODE_1_length_588386_cov_32.551147 588386 562922 563163 1283 1241 68
TFA_p7.G11 546 84 546 - NODE_1_length_588386_cov_32.551147 588386 578993 571455 462 68
HTF_p21.F9 1245 639 1221 - NODE_1_length_588386_cov_32.551147 588386 569178 569752 563 582 68
HTF_p5.D5 1803 1193 1779 - NODE_1_length_588386_cov_32.551147 588386 569178 569756 566 586 68
HTF_p25.A12 1383 565 1359 - NODE_1_length_588386_cov_32.551147 588386 569178 569964 765 794 68
HTF_p7.G11 1830 9 1796 - NODE_1_length_588386_cov_32.551147 588386 569188 578967 1747 1787 68

i282      zd:i:2  rl:i:0  cg:Z:123M123I1369M48I4930M
i288      zd:i:1  rl:i:0  cg:Z:5942M
rl:i:0    cg:Z:1241M
cg:Z:462M
cg:Z:582M
cg:Z:586M
cg:Z:794M
rl:i:0    cg:Z:1787M
```

also p7.G11

check all rest are p7.G11 haplotypes, with length 1830:

PG32

TFA_p7.G11	546	84	546
HTF_p21.F9	1245	639	1245
HTF_p25.C2	1803	1207	1803
HTF_p5.D5	1803	1193	1803
HTF_p25.A12	1383	565	1383
HTF p7.G11	1830	9	1830

PG1: no taillocin

PG14:

TFA_p7.G11	546	84	546
HTF_p21.F9	1245	639	1221
HTF_p25.C2	1803	1207	1779
HTF_p5.D5	1803	1193	1779
HTF_p25.A12	1383	565	1359
HTF_p7.G11	1830	9	1796

PG39:

TFA_p7.G11	546	84	546
HTF_p21.F9	1245	639	1221
HTF_p5.D5	1803	1193	1779
HTF_p25.A12	1383	565	1359
HTF_p7.G11	1830	9	1796

later do MSA of tail fiber p7.G11 haplotypes TFA/HTF.

for now focus on tape measure:

cds gene seqs:

```
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```

```
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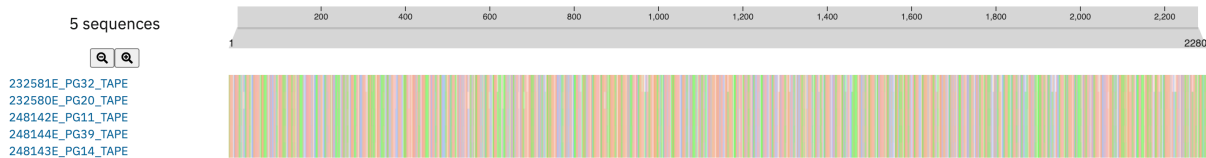
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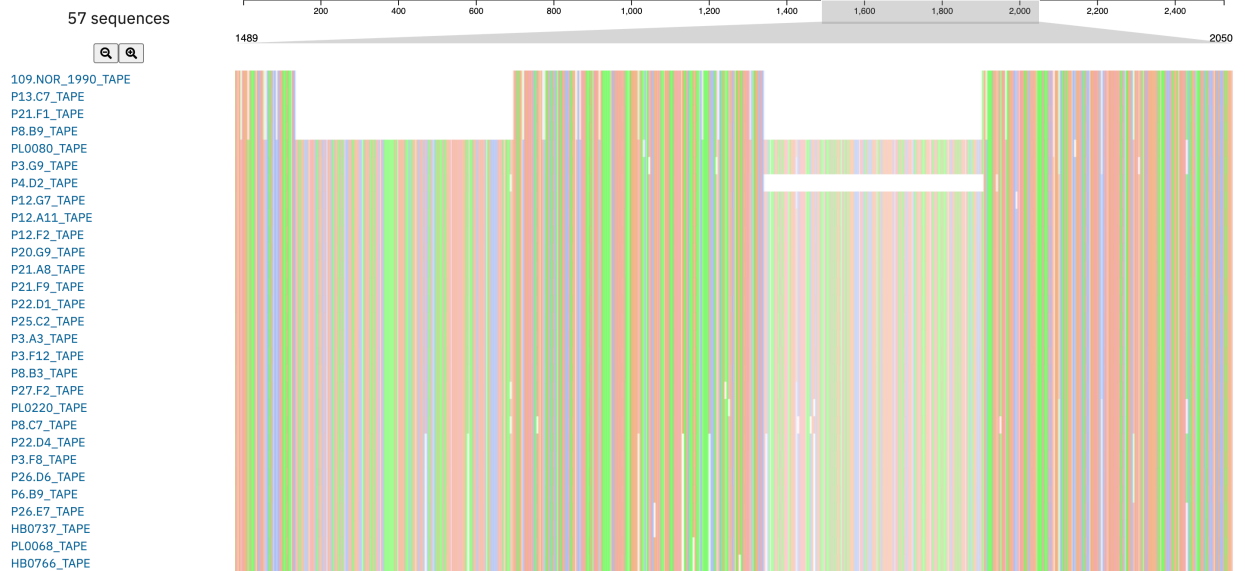
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aa translated:

eg:

Translate is a tool which allows the translation of a nucleotide (DNA/RNA) sequence to a protein sequence.

DNA or RNA sequence

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Output format

☐ Verbose: Met, Stop, spaces between residues

☒ Compact: M, -, no spaces

☐ Includes nucleotide sequence

☐ Includes nucleotide sequence, no spaces

Genetic codes - [See NCBI's genetic codes](#)

Standard

[reset](#) [TRANSLATE!](#)

Results of translation

- [Open reading frames](#) are highlighted in red
- Select your initiator on one of the following frames to retrieve your amino acid sequence

[Download all the translated frames](#)

5' Frame 1

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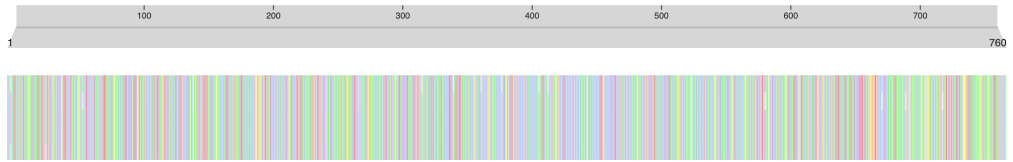
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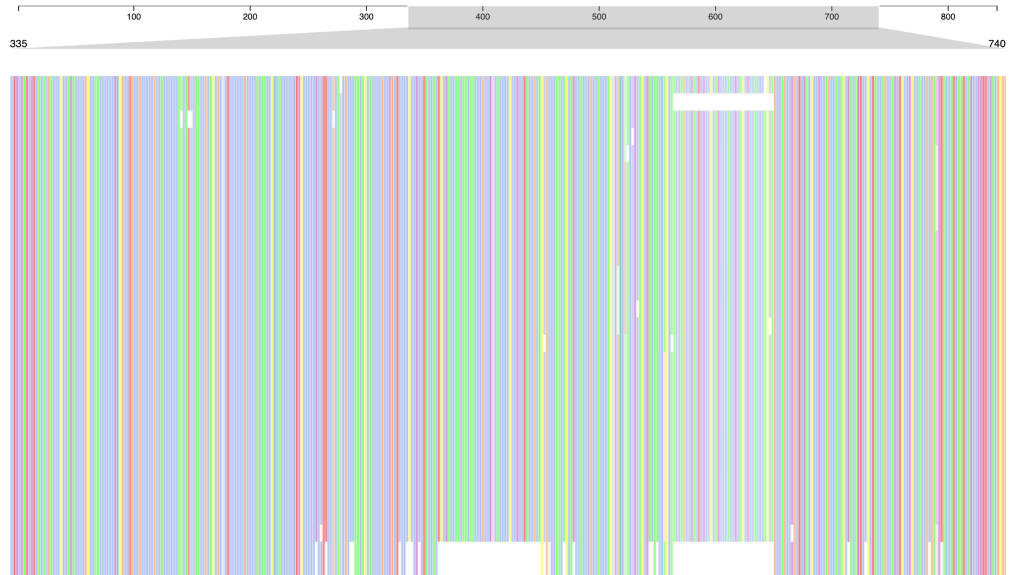


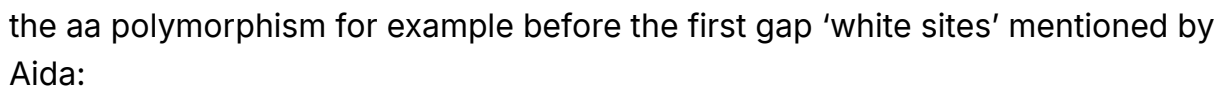
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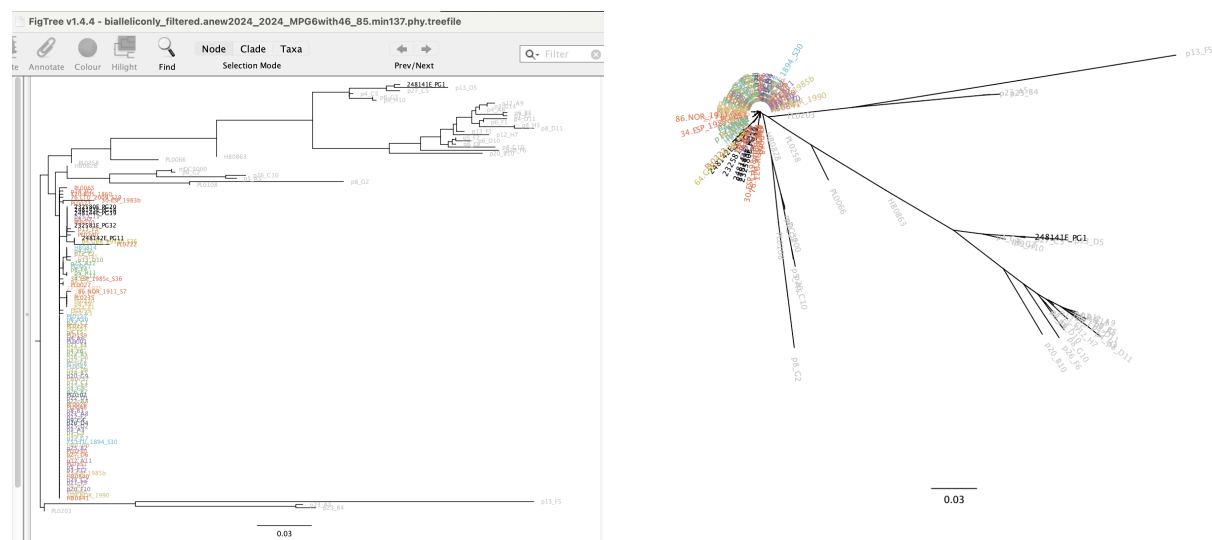


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allmergedSNPs,nomultiSNPs,biallelicSNPs

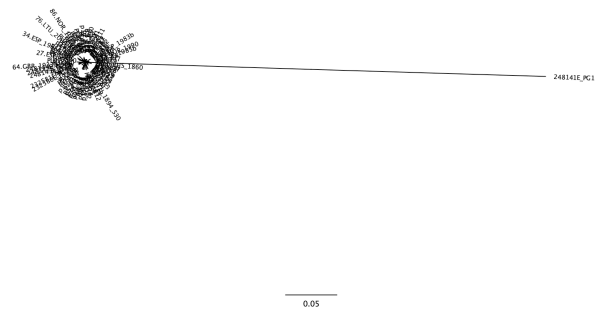
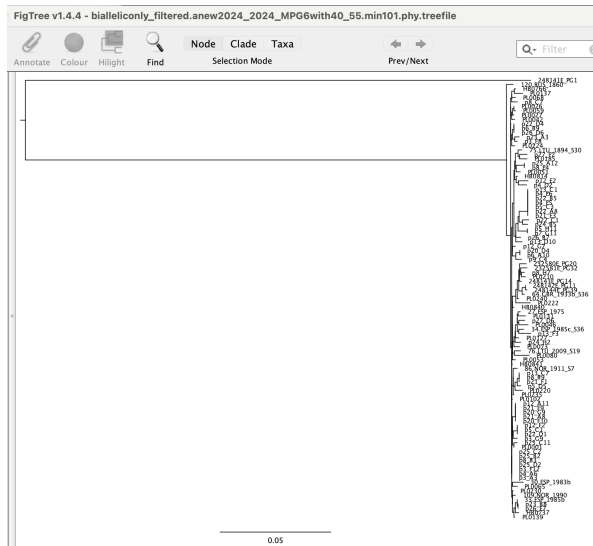
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allmergedSNPs,nomultiSNPs,biallelicSNPs

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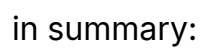


MPG5 with 40 55 OTU5:

total 100

allmergedSNPs,nomultiSNPs,biallelicSNPs

460039 459266 20131



1. Among the 6 MPGs, 5 pathogens are within OTU5 while the commensal is out with nonOTU5 (in the trees of: 6 MPGs with 95 OTU5 and 30 nonOTU5 (biSNP: 280) and 6 MPGs with 95 OTU5 (biSNP: 762)).
2. when assigned with 95 OTU5, the 5 MPG pathogens clustered together in one clade (in black; bi-SNP 20,131).